

Geodesic Newton Direction for the Extended Embedding

1 Setup

Consider the extended embedding of a conic program with cone $\mathcal{C} = \mathbb{R}_+^n$:

$$\begin{aligned} \min \quad & \alpha\theta \\ \text{s.t.} \quad & Ax - b\tau = r_p\theta \end{aligned} \tag{1}$$

$$-A^T y - s + c\tau = r_d\theta \tag{2}$$

$$\langle b, y \rangle - \langle c, x \rangle - \kappa = r_g\theta \tag{3}$$

$$\langle r_p, y \rangle + \langle r_d, x \rangle + r_g\tau = -\alpha \tag{4}$$

$$x \in \mathbb{R}_+^n, s \in \mathbb{R}_+^n, \tau \geq 0, \kappa \geq 0$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, and $y \in \mathbb{R}^m$, $\theta \in \mathbb{R}$ are free variables.

2 Self-duality

The embedding is self-dual: if $(x, s, y, \tau, \kappa, \theta)$ is feasible, then swapping

$$(x, s, \tau, \kappa, y) \longleftrightarrow (s, x, \kappa, \tau, -y)$$

gives a dual-feasible point. This symmetry is the key structural property.

3 Geodesic parametrization

For the product cone $\mathbb{R}_+^n \times \mathbb{R}_+^n \times \mathbb{R}_+ \times \mathbb{R}_+$ with rank $R = 2n + 2$, the geodesic parametrization at scaling $W = \text{diag}(w_x, w_s, w_\tau, w_\kappa)$ and barrier parameter $k = 1/\sqrt{\mu}$ gives:

$$\text{slack}_i = \frac{1 - d_i}{k W_i}, \tag{5}$$

$$\lambda_i = k W_i (1 + d_i). \tag{6}$$

For the $x \geq 0$ constraint, the “slack” is x itself and λ_x is its dual. For $s \geq 0$, the slack is s and λ_s is its dual.

4 Self-duality implies $d_x = -d_s$

By self-duality, x is simultaneously:

1. The slack of the $x \geq 0$ constraint: $x_i = \frac{1-d_{x,i}}{k w_{x,i}}$.
2. The dual variable of the $s \geq 0$ constraint: $x_i = k w_{s,i} (1 + d_{s,i})$.

Symmetrically, s is:

1. The slack of $s \geq 0$: $s_i = \frac{1-d_{s,i}}{k w_{s,i}}$.
2. The dual of $x \geq 0$: $s_i = k w_{x,i} (1 + d_{x,i})$.

4.1 Nesterov–Todd scaling

Impose the NT scaling $w_{x,i} = w_{s,i}^{-1}$. Then from identifications (1) and (2) for x :

$$\frac{1-d_{x,i}}{k w_{x,i}} = k w_{s,i} (1 + d_{s,i}) = k w_{x,i}^{-1} (1 + d_{s,i}).$$

Multiply both sides by $k w_{x,i}$:

$$1 - d_{x,i} = k^2 (1 + d_{s,i}).$$

Similarly from s :

$$\frac{1-d_{s,i}}{k w_{s,i}} = k w_{x,i} (1 + d_{x,i}).$$

Multiply by $k w_{s,i} = k w_{x,i}^{-1}$:

$$1 - d_{s,i} = k^2 (1 + d_{x,i}).$$

Adding both equations:

$$(1 - d_{x,i}) + (1 - d_{s,i}) = k^2 [(1 + d_{s,i}) + (1 + d_{x,i})].$$

$$2 - (d_{x,i} + d_{s,i}) = k^2 [2 + (d_{x,i} + d_{s,i})].$$

Let $\sigma = d_{x,i} + d_{s,i}$:

$$2 - \sigma = k^2 (2 + \sigma) \implies \sigma (1 + k^2) = 2(1 - k^2) \implies \sigma = \frac{2(1 - k^2)}{1 + k^2}.$$

From the first equation alone:

$$1 - d_{x,i} = k^2 (1 + d_{s,i}) \implies d_{x,i} = 1 - k^2 - k^2 d_{s,i}.$$

From the second:

$$d_{s,i} = 1 - k^2 - k^2 d_{x,i}.$$

Substituting:

$$d_{s,i} = 1 - k^2 - k^2(1 - k^2 - k^2 d_{s,i}) = (1 - k^2)(1 - k^2) + k^4 d_{s,i}.$$

$$d_{s,i}(1 - k^4) = (1 - k^2)^2 \implies d_{s,i} = \frac{(1 - k^2)^2}{(1 - k^2)(1 + k^2)} = \frac{1 - k^2}{1 + k^2}.$$

This gives $d_{s,i}$ as a *constant* (independent of i), which is too strong. The issue is that identifying x simultaneously as slack of $x \geq 0$ and dual of $s \geq 0$ over-constrains the system at general k .

4.2 Correct identification at $k = 1$

At $k = 1$ (the initial barrier), the equations give $\sigma = 0$, so $d_{x,i} = -d_{s,i}$. More generally, the self-dual structure means the central path satisfies $x \circ s = \mu e$ and $\tau \kappa = \mu$, which in the geodesic parametrization becomes $d_x = -d_s$ and $d_\tau = -d_\kappa$ **at every point on the central path**.

To see this directly: on the central path,

$$x_i s_i = \mu, \quad \text{i.e.,} \quad \frac{(1 - d_{x,i})(1 - d_{s,i})}{k^2 w_{x,i} w_{s,i}} \cdot k^2 w_{x,i} w_{s,i} = \frac{(1 - d_{x,i})(1 + d_{x,i})}{1} = \mu$$

Wait — this is x_i as slack times $\lambda_{x,i}$ as dual of the same constraint:

$$x_i \cdot \lambda_{x,i} = \frac{1 - d_{x,i}}{k w_{x,i}} \cdot k w_{x,i} (1 + d_{x,i}) = (1 - d_{x,i})(1 + d_{x,i}) = 1 - d_{x,i}^2 = \mu.$$

This gives $d_{x,i}^2 = 1 - \mu$ for each i on the central path, but doesn't directly relate d_x to d_s .

4.3 Direct argument from the Newton equations

The clearest argument: the Newton direction at (W, k) solves a linear system. By the self-duality of the embedding, the solution inherits the symmetry. Specifically, if we apply the primal-dual swap $(x, s, \tau, \kappa) \rightarrow (s, x, \kappa, \tau)$ and $y \rightarrow -y$ to the Newton equations, and if W satisfies $w_x = w_s^{-1}$ and $w_\tau = w_\kappa^{-1}$ (NT scaling for both pairs), then the transformed direction $(\tilde{d}_x, \tilde{d}_s) = (-d_s, -d_x)$ also solves the Newton equations. By uniqueness:

$$\boxed{d_x = -d_s, \quad d_\tau = -d_\kappa.}$$

5 Implications for the KKT system

5.1 Reduced dimension

With $d_s = -d_x$ and $d_\kappa = -d_\tau$, the direction has $n + 1$ independent cone components instead of $2n + 2$. The effective rank is

$$R_{\text{eff}} = n + 1.$$

5.2 Elimination of s and κ

Since $d_{s,i} = -d_{x,i}$, from (5):

$$s_i = \frac{1 - d_{s,i}}{kw_{s,i}} = \frac{1 + d_{x,i}}{kw_{x,i}^{-1}} = kw_{x,i}(1 + d_{x,i}) = \lambda_{x,i}.$$

So $s = \lambda_x$: the dual slack equals the dual variable of $x \geq 0$. Similarly $\kappa = \lambda_\tau$.

Substituting into the dual equality (2):

$$-A^T y - \lambda_x + c\tau = r_d \theta.$$

This is the standard dual feasibility condition with s eliminated.

5.3 Elimination of d

The direction d_x is not an independent variable — it is determined by the primal variables through (??):

$$d_{x,i} = 1 - w_{x,i} \cdot x_i.$$

Substituting into the expressions for the dual:

$$\lambda_{x,i} = k w_{x,i}(1 + d_{x,i}) = k w_{x,i}(2 - w_{x,i}x_i) = 2k w_{x,i} - k w_{x,i}^2 x_i.$$

Since $s_i = \lambda_{x,i}$:

$$s_i = 2k w_{x,i} - k w_{x,i}^2 x_i. \quad (7)$$

Similarly:

$$\kappa = 2k w_\tau - k w_\tau^2 \tau. \quad (8)$$

5.4 Reduced system in (x, y, τ, θ)

Substituting (7) and (8) into the equality constraints (1)–(4), we eliminate s , κ , and d entirely. The independent variables are

$$(x, y, \tau, \theta) \in \mathbb{R}^{n+m+2}$$

with cone constraints $x \geq 0$, $\tau \geq 0$ (rank $n + 1$).

Equality (1) (unchanged):

$$Ax - b\tau = r_p \theta. \quad (\text{E1})$$

Equality (2) with s from (7):

$$\begin{aligned} -A^T y - (2kw_x - kW_x^2 x) + c\tau &= r_d \theta \\ kW_x^2 x - A^T y + c\tau &= r_d \theta + 2kw_x. \end{aligned} \quad (\text{E2})$$

Equality (3) with κ from (8):

$$\begin{aligned}\langle b, y \rangle - \langle c, x \rangle - (2kw_\tau - kw_\tau^2\tau) &= r_g\theta \\ -\langle c, x \rangle + \langle b, y \rangle + kw_\tau^2\tau &= r_g\theta + 2kw_\tau.\end{aligned}\tag{E3}$$

Equality (4) (unchanged — involves only x, y, τ):

$$\langle r_d, x \rangle + \langle r_p, y \rangle + r_g\tau = -\alpha.\tag{E4}$$

5.5 KKT system

Note that (E2) already contains the term kW_x^2x , which is k times the x -block of the Gram matrix applied to x . Similarly (E3) contains $kw_\tau^2\tau$. This means the equality constraints absorb the Gram contributions at barrier parameter k .

The constraint Jacobian J (rows = constraints, columns = variables (x, y, τ, θ)) is:

$$J = \begin{pmatrix} A & 0 & -b & -r_p \\ kW_x^2 & -A^T & c & r_d \\ -c^T & b^T & kw_\tau^2 & -r_g \\ r_d^T & r_p^T & r_g & 0 \end{pmatrix}.$$

The Gram matrix from the cone constraints $x \geq 0, \tau \geq 0$ is

$$G = \text{diag}(W_x^2, 0_m, w_\tau^2, 0).$$

The full KKT system $\begin{pmatrix} G & J^T \\ J & 0 \end{pmatrix} \begin{pmatrix} z \\ \nu \end{pmatrix} = \begin{pmatrix} g \\ h \end{pmatrix}$ is:

$$\begin{pmatrix} W_x^2 & & & & A^T & kW_x^2 & -c & r_d \\ & 0 & & & 0 & -A & b & r_p \\ & & w_\tau^2 & & -b^T & c^T & kw_\tau^2 & r_g \\ & & & 0 & -r_p^T & r_d^T & -r_g & 0 \\ A & 0 & -b & -r_p & & & & \\ kW_x^2 & -A^T & c & r_d & & & & \\ -c^T & b^T & kw_\tau^2 & -r_g & & & & \\ r_d^T & r_p^T & r_g & 0 & & & & \end{pmatrix} \begin{pmatrix} x \\ y \\ \tau \\ \theta \\ \nu_p \\ \nu_d \\ \nu_g \\ \nu_n \end{pmatrix} = \begin{pmatrix} 2w_x \\ 0 \\ 2w_\tau \\ -k\alpha \\ 0 \\ 2kw_x \\ 2kw_\tau \\ -\alpha \end{pmatrix}\tag{9}$$

where $W_x = \text{diag}(w_x)$, $\nu = (\nu_p, \nu_d, \nu_g, \nu_n)$ are the duals of constraints (E1)–(E4), and the RHS incorporates the $2kw_x$ and $2kw_\tau$ terms from the substitution of s and κ .

Note: the top-right block is J^T and the bottom-left is J , as required by KKT symmetry.

6 Elimination of x

The cone variable x has invertible Gram block W_x^2 . From the stationarity condition (row 1 of (9)):

$$W_x^2x + A^T\nu_p + kW_x^2\nu_d - c\nu_g + r_d\nu_n = 2w_x.\tag{10}$$

Solving for x with $D = W_x^{-2}$:

$$x = 2W_x^{-1} - DA^T\nu_p - k\nu_d + Dc\nu_g - Dr_d\nu_n. \quad (11)$$

6.1 Substitution into constraints

Substitute (11) into each constraint row.

(E1): $Ax - b\tau - r_p\theta = 0.$

$$\begin{aligned} & A(2W_x^{-1} - DA^T\nu_p - k\nu_d + Dc\nu_g - Dr_d\nu_n) - b\tau - r_p\theta = 0 \\ \Rightarrow & -ADA^T\nu_p - kA\nu_d + ADc\nu_g - ADr_d\nu_n - b\tau - r_p\theta = -2AW_x^{-1}. \end{aligned} \quad (12)$$

(E2): $kW_x^2x - A^Ty + c\tau + r_d\theta = 2kw_x.$ Since $kW_x^2 \cdot D = kI$ and $kW_x^2 \cdot W_x^{-1} = kw_x$:

$$\begin{aligned} & 2kw_x - kA^T\nu_p - k^2W_x^2\nu_d + kc\nu_g - kr_d\nu_n - A^Ty + c\tau + r_d\theta = 2kw_x \\ \Rightarrow & -kA^T\nu_p - k^2W_x^2\nu_d + kc\nu_g - kr_d\nu_n - A^Ty + c\tau + r_d\theta = 0. \end{aligned} \quad (13)$$

The constant terms cancel.

(E3): $-c^Tx + b^Ty + kw_\tau^2\tau - r_g\theta = 2kw_\tau.$

$$\begin{aligned} & -c^T(2W_x^{-1} - DA^T\nu_p - k\nu_d + Dc\nu_g - Dr_d\nu_n) + b^Ty + kw_\tau^2\tau - r_g\theta = 2kw_\tau \\ \Rightarrow & c^TDA^T\nu_p + kc^T\nu_d - c^TDc\nu_g + c^TDr_d\nu_n + b^Ty + kw_\tau^2\tau - r_g\theta = 2kw_\tau + 2c^TW_x^{-1}. \end{aligned} \quad (14)$$

(E4): $r_d^Tx + r_p^Ty + r_g\tau = -\alpha.$

$$\Rightarrow -r_d^TDA^T\nu_p - kr_d^T\nu_d + r_d^TDc\nu_g - r_d^TDr_d\nu_n + r_p^Ty + r_g\tau = -\alpha - 2r_d^TW_x^{-1}. \quad (15)$$

6.2 Reduced system

The remaining stationarity conditions (rows 2–4 of (9)) do not involve x :

$$\text{row 2 (y): } -A\nu_d + b\nu_g + r_p\nu_n = 0, \quad (16)$$

$$\text{row 3 (\tau): } w_\tau^2\tau - b^T\nu_p + c^T\nu_d + kw_\tau^2\nu_g + r_g\nu_n = 2w_\tau, \quad (17)$$

$$\text{row 4 (\theta): } -r_p^T\nu_p + r_d^T\nu_d - r_g\nu_g = -k\alpha. \quad (18)$$

Equations (12)–(15) and (16)–(18) form a system in $(y, \tau, \theta, \nu_p, \nu_d, \nu_g, \nu_n)$ of size $2m + n + 3$.

Define the *Schur complement matrix* $S = ADA^T = AW_x^{-2}A^T \in \mathbb{R}^{m \times m}$ (symmetric positive semidefinite). The constraint block after elimination is:

$$\begin{pmatrix} 0 & 0 & 0 & 0 & -A & b & r_p \\ 0 & w_\tau^2 & 0 & -b^T & c^T & kw_\tau^2 & r_g \\ 0 & 0 & 0 & -r_p^T & r_d^T & -r_g & 0 \\ 0 & -b & -r_p & -S & -kA & ADc & -ADr_d \\ -A^T & c & r_d & -kA^T & -k^2W_x^2 & kc & -kr_d \\ b^T & kw_\tau^2 & -r_g & c^TDA^T & kc^T & -c^TDc & c^TD r_d \\ r_p^T & r_g & 0 & -r_d^TDA^T & -kr_d^T & r_d^TDc & -r_d^TD r_d \end{pmatrix} \begin{pmatrix} y \\ \tau \\ \theta \\ \nu_p \\ \nu_d \\ \nu_g \\ \nu_n \end{pmatrix} = \begin{pmatrix} 0 \\ 2w_\tau \\ -k\alpha \\ -2AW_x^{-1} \\ 0 \\ 2kw_\tau + 2c^TW_x^{-1} \\ -\alpha - 2r_d^TW_x^{-1} \end{pmatrix}. \quad (19)$$

Key observation. The expensive block is $S = AW_x^{-2}A^T$ ($m \times m$), which is exactly the normal-equations matrix from standard LP interior-point methods. When $m \ll n$, this is a significant reduction: we have eliminated n variables (x) at the cost of a dense $m \times m$ Schur complement.

6.3 Dimensions

	Full embedding	Reduced (NT)	x -eliminated
Primal variables	$2n + m + 3$	$n + m + 2$	$m + 2$
Equality duals	$m + n + 2$	$m + n + 2$	$m + n + 2$
Total system size	$3n + 2m + 5$	$2n + 2m + 4$	$2m + n + 3$
Cone rank (RowSpace)	$2n + 2$	$n + 1$	1 (τ only)
Dense block	—	—	$S = AW_x^{-2}A^T$ ($m \times m$)

For the original LP with m constraints and n variables, the x -eliminated system has size $2m + n + 3$, which is smaller than the NT-reduced system ($2n + 2m + 4$) whenever $n > 1$. The dominant cost is forming and factoring the Schur complement $S \in \mathbb{R}^{m \times m}$.

6.4 Recovery

Given the solution $(y^*, \tau^*, \theta^*, \nu^*)$ of (19), recover x from (11):

$$x^* = 2W_x^{-1} - DA^T\nu_p^* - k\nu_d^* + Dc\nu_g^* - Dr_d\nu_n^*.$$

Then from $(x^*, y^*, \tau^*, \theta^*)$, recover:

$$d_{x,i} = 1 - w_{x,i}x_i^*, \quad (20)$$

$$d_\tau = 1 - w_\tau\tau^*, \quad (21)$$

$$s_i^* = 2kw_{x,i} - kw_{x,i}^2x_i^* = kw_{x,i}(2 - w_{x,i}x_i^*) = kw_{x,i}(1 + d_{x,i}), \quad (22)$$

$$\kappa^* = kw_\tau(1 + d_\tau), \quad (23)$$

$$d_{s,i} = -d_{x,i}, \quad d_\kappa = -d_\tau. \quad (24)$$

The complementarity gap is

$$\mu((n+1) - \|d_x\|^2 - d_\tau^2) = \frac{1}{k^2}((n+1) - \|d_x\|^2 - d_\tau^2).$$

7 Structure of the θ update

7.1 θ as duality gap residual

From the gap constraint (E3) before eliminating s and κ :

$$\theta = \frac{\langle b, y \rangle - \langle c, x \rangle - \kappa}{r_g}. \quad (25)$$

So θ measures the *residual duality gap*: the quantity $\langle b, y \rangle - \langle c, x \rangle - \kappa$ is exactly $r_g \theta$. At the fixed point ($\theta = 1$), this recovers the definition of r_g . At optimality ($\theta = 0$), we get $\langle b, y \rangle = \langle c, x \rangle + \kappa$, and if $\tau > 0$ with $\kappa \rightarrow 0$, this yields strong duality for the original LP.

Similarly, the primal and dual residuals are proportional to θ :

$$Ax - b\tau = r_p \theta \quad (\text{primal residual}), \quad (26)$$

$$-A^T y - s + c\tau = r_d \theta \quad (\text{dual residual}). \quad (27)$$

All three residuals vanish simultaneously as $\theta \rightarrow 0$.

7.2 θ -free equations

In the x -eliminated system (19), θ appears in rows 4–6 (the eliminated E1–E3) but **not** in:

- Row 1: y -stationarity (16),
- Row 2: τ -stationarity (17),
- Row 3: θ -stationarity (18),
- Row 7: the normalization constraint (E4) (15).

These four equations constrain (y, τ, ν) independently of θ . Once (y, τ, ν) are determined, θ is recovered from any of E1–E3; the scalar E3 is most convenient:

$$\theta = \frac{1}{r_g} \left(c^T D A^T \nu_p + k c^T \nu_d - c^T D c \nu_g + c^T D r_d \nu_n + b^T y + k w_\tau^2 \tau - 2 k w_\tau - 2 c^T W_x^{-1} \right). \quad (28)$$

7.3 Predictor step from a centered point

At a centered point ($d_x = 0, d_\tau = 0$), the Newton step produces a direction $(\delta d_x, \delta d_\tau, \delta \theta)$. The geodesic parametrization requires $|d_{x,i} + \delta d_{x,i}| < 1$ and $|d_\tau + \delta d_\tau| < 1$, so from centering: $|\delta d_{x,i}| < 1, |\delta d_\tau| < 1$.

The cone rank is $R = n + 1$ (from $x \in \mathbb{R}_+^n$ and $\tau \geq 0$). By the standard self-dual IPM analysis, the predictor direction satisfies

$$\|\delta d\|^2 = \sum_i (\delta d_{x,i})^2 + (\delta d_\tau)^2 \propto (\delta \theta / \theta)^2 \cdot R.$$

A safe step (keeping $\|d\|_\infty < 1$) yields

$$|\delta \theta| \leq \frac{\theta}{\sqrt{R}} = \frac{\theta}{\sqrt{n+1}},$$

giving the update

$$\boxed{\theta_{\text{new}} \approx \theta \left(1 - \frac{1}{\sqrt{n+1}} \right)}. \quad (29)$$

This is the standard $O(\sqrt{n} \log(1/\varepsilon))$ convergence rate for self-dual embeddings, with the effective cone rank $n + 1$ (not the original $2n + 2$) thanks to the NT reduction $d_x = -d_s$.